

| Lectures                       | Lecture Topics                                                                                                                                   | Lab                                                                                                                                    |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Week 1</b><br>1/13<br>1/15  | Introduction and Expectations<br>Introduction to Simulink and dSPACE<br>Sensors <ul style="list-style-type: none"> <li>Accelerometers</li> </ul> | <b>Lab 1:</b> Room 142 Safety Lecture. Introduction to data acquisition with Simulink and dSPACE.                                      |
| <b>Week 2</b><br>1/20<br>1/22  | Data Acquisition of Dynamic Signals                                                                                                              | <b>Lab 2:</b> Room 142 Three axis accelerometer.                                                                                       |
| <b>Week 3</b><br>1/27<br>1/29  | MEMs Gyroscope Sensors<br>EOM for Quadcopter Arm<br>Angle Measurement Uncertainty<br>Using Accelerometer                                         | <b>Lab 3:</b> Room 142 Acquire two axis data from MEMs gyroscope.                                                                      |
| <b>Week 4</b><br>2/3<br>2/5    | Actuators<br>Pulse Width Modulated PWM<br>Brushed Motors<br>Hall Effect Sensors                                                                  | <b>Lab 4:</b> Room 142 PWM actuation of motor and servo. Hall effect sensor and spectrum analysis.                                     |
| <b>Week 5</b><br>2/10<br>2/12  | Power Spectrum<br>Aliasing in Spectra<br>Kalman Filtering Basics                                                                                 | <b>Lab 5:</b> Room 142 System identification of a R/C servo using an accelerometer sensor.                                             |
| <b>Week 6</b><br>2/17<br>2/19  | System Identification<br>Open Loop Control of Brushless Motors                                                                                   | <b>Lab 6a:</b> Room 142 Open loop control of quadcopter arm.<br><b>Lab report 1 on this lab due at lab time week of March 2.</b>       |
| <b>Week 7</b><br>2/24<br>2/26  | Closed Loop Control PID                                                                                                                          | <b>Lab 6b:</b> Room 142 Open loop control of quadcopter arm. Complete if needed.                                                       |
| <b>Week 8</b><br>3/3<br>3/5    | Midterm Review                                                                                                                                   | <b>Lab 7:</b> Room 142 Open lab time to practice for the practical exam.<br><b>Lab report-1 due at lab time.</b>                       |
| <b>Week 9</b><br>3/10<br>3/12  | <b>3/10 - Exam 1 Written part (40% of Exam 1 total)</b><br><br>3/12 Time Varying Signal Analysis                                                 | <b>Exam 1: Room 142 Practical part (60% of Exam 1 total).</b>                                                                          |
| <b>Week 10</b><br>3/17<br>3/19 | No Lectures Due to Spring Break                                                                                                                  | No labs due to spring break.                                                                                                           |
| <b>Week 11</b><br>3/24<br>3/26 | Closed Loop Control PID                                                                                                                          | <b>Lab 8:</b> Room 142 PID. Closed loop control of quadcopter arm.<br><b>Lab report 2 on this lab due at lab time week of April 6.</b> |

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| <b>Week 12</b><br>3/31<br>4/2  | Design of Experiments.                                       | <b>Lab 9:</b> Room 142 System identification of quadcopter arm.                                    |
| <b>Week 13</b><br>4/7<br>4/9   | Digital Sensors and Raspberry Pi<br>Fluid Power Fundamentals | <b>Lab 10a:</b> Room 142 Digital sensors and Raspberry Pi.<br><b>Lab report 2 due at lab time.</b> |
| <b>Week 14</b><br>4/14<br>4/16 | Fluid Power Motors and Actuators                             | <b>Lab 10b:</b> Room 142 Digital sensors and Raspberry Pi. Complete if needed.                     |
| <b>Week 15</b><br>4/21<br>4/23 | Fluid Power Control                                          | <b>Lab 11:</b> Room 142 Fluid power trainer.                                                       |
| <b>Week 16</b><br>4/28<br>4/30 | Course Review                                                | No labs.                                                                                           |
| <b>Final Exam Week</b><br>5/4  | <b>Exam 2 - Location and Time to be Announced</b>            | <b>No labs.</b><br><b>Lab notebook turned in for a grade.</b>                                      |

Also, by appointment.

**Required Textbook:** None.

**Grades:**

- Based on a curve.
- Lab reports – 30%
- Homework – 10%
- Attendance and participation in labs – 10%
- Quality of laboratory notebook - 10%
- Exam 1 – 20% (12% practical 8% written)
- Exam 2 – 20% (written only)

**Labs:**

Participation in labs counts toward 10% of the grade. Just like the lecture, labs will start on time and if the student arrives late, it will be the responsibility of the student to catch up.

If the student misses their regularly scheduled lab session, then the student must make arrangements with the instructor and teaching assistant to complete the lab that week. If the lab is not completed that week, the student cannot turn in a lab report (where applicable) and in addition the final grade will be reduced by one letter grade for each lab missed.

If the student cannot attend a lab for the entire week due to a medical reason or other approved absence, it is the responsibility of the student to arrange a review of the lab with the instructor or teaching assistant. After the review, the student must obtain the lab data from one of their lab partners and write up the lab in their lab notebook as required. The student will complete these activities before the date the lab notebook is required to be turned in for grading.

The grade for lab reports turned in late will be reduced by 20%. Late lab reports will no longer be accepted for grading one week after the due date.

Lab notebooks must be turned in with the second exam. No exceptions.

Each student will have access to an equipment cart for use during the semester. Students are responsible for the care and well-being of the equipment on the cart. It will be the responsibility of the student to notify the instructor or teaching assistant of any missing or damaged equipment from the cart. Carts will be self-assigned in groups of two (minimum) or four (maximum).

Lab notebooks will be provided to the students. Each student must keep a detailed lab notebook containing all information collected during the labs, which will be turned in at the end of Exam 2.

For each lab, the student will need to record:

- Date of the experiment
- Name of experiment and Goal of experiment (Use multiple sections if the lab involves multiple parts)
- List of teammates
- Experimental setup
  - Equipment used for the work (e.g. BK Precision 2704c multimeter...)
  - Relevant, data acquisition parameters (number of samples, acquisition rate, etc....)
  - Sketch of experiment with clear labels; photographs are acceptable
  - Procedure including Simulink models (where applicable)
- Summary of results
  - Table of measurements / graphs with titles
  - Brief discussion and conclusions
- Any piece of information (e.g. printed graphs, tables) in your lab notebook must be taped.
- Answers to any questions that are listed in the lab handouts.

The lab notebook will not be as detailed as a lab report but needs to contain sufficient information for someone else to closely reproduce your experiments.

The student is allowed to use the lab notebook for Exam 1 (written and practical parts) and Exam 2. Fill out the lab notebook as you go during lab time. Don't wait until later to start writing your notes, you won't remember what the equipment was, or what parameters you used.

Two lab reports are due throughout the semester. The lab report format and further information will be given at the appropriate time.

### **Exams:**

No make-up exams will be given except for emergency situations. Contact the [REDACTED] reports (where applicable).

### **Homework:**

Homework will be assigned at various times throughout the semester. Each student must turn in a unique homework assignment on paper. Electronic versions of the homework will not be accepted. Tablet printouts will be accepted.

Homework assignments will be due usually at lab time. Homework assignments turned in after the lab session will be considered late. The score for a late homework assignment will be reduced by 20%. Homework assignments will no longer be accepted one week past the due date.

### **ADA Statement:**

Reasonable accommodation will be provided for students with documented disabilities. In order to receive accommodations, students must obtain a letter of accommodation from the Center for Disability Resources. The Center for [REDACTED]

located at <https://www.iit.edu/cdr/services/reasonable-accommodations>.

### **Academic Honesty and Collaboration Policy:**

IIT expects students to maintain high standards of academic integrity. Students preparing for the practice of a profession are expected to conform to a code of integrity and ethical standards commensurate with the high expectations society places on practitioners of a learned profession. No student may seek to gain an unfair advantage over another. The Code of Academic Honesty is explained in the IIT Student Handbook, and all students are expected to know and adhere to this code.

Departmental policies on academic honesty are consistent with those of the University. The Department policies are provided at [https://www.iit.edu/sites/default/files/2021-01/honesty\\_policy\\_-\\_mmae.pdf](https://www.iit.edu/sites/default/files/2021-01/honesty_policy_-_mmae.pdf).

Collaboration on homework is not allowed.

Collaboration on the laboratory notebooks and laboratory reports is not allowed. Each student should write their own lab reports, and, on their own, analyze and plot the data acquired in lab. Plagiarism, which is “an act or instance of using or closely imitating the language and thoughts of another author without authorization and the representation of that author's work as one's own, as by not crediting the original author” is not allowed. Definition from dictionary.com.

Collaboration, plagiarism or the use of artificial intelligence (AI) on a lab report will result in a zero grade for the lab report. A repeated violation will result in the lowering of the final grade by one letter grade.

Collaboration on Exam 1 and 2 is not allowed.

Infractions of the Honesty and Collaboration policy will be referred to the Academic Honesty Committee.

### **Classroom Expectations Regarding Diversity, Equity, and Inclusion:**

The MMAE Department statement on Diversity, Equity, and Inclusion including the incident reporting process is found in the [MMAE Code of Conduct](#).

It is expected everyone will treat each other with empathy and civility both in person and online.